The cost of one measured ecbench child

2026-10-24 · `research/ecbench_child_overhead_20261024/` · measured: Callgrind instruction counts; wall time descriptive (shared 4-vCPU Xeon VM, L0, no PMU) · class: **engineering** (GAE unchanged, solves identical across binaries)

Every measured run of every ecbench session is a fresh `ecbench exec` process. Arms: before 2184 (`1b74250e9`), after its child-path commits (`0f5d9c272`), `main` (`004b792b7`), and this PR (`645666691`). This PR removes the alias-table parse that `main` had grown into every index-calculus child. Inputs: the first 18 measured executions of `docs/ecbench/specs/smoke-prime.json` (20-bit prime-order curve, every method). Protocol: `PROTOCOL.md`, frozen before timing.

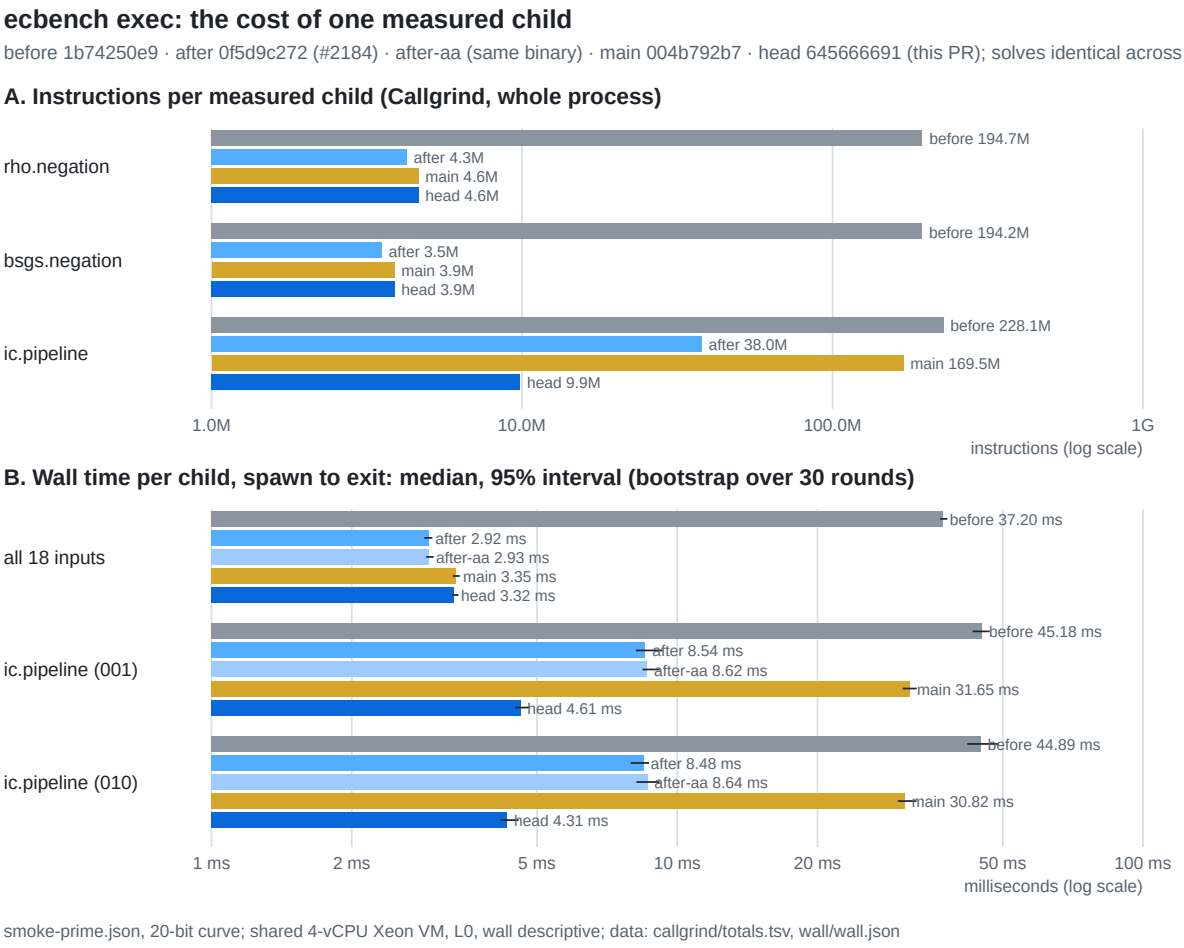


Figure 1: Instructions (Callgrind, whole process, every dump part) and wall time per child. Data: `callgrind/totals.tsv`, `wall/wall.json`.

Instructions per child

method (input)	before	after #2184	`main`	this PR
`rho.negation` (008)	194.7M	4.26M	4.64M	4.64M
`bsgs.negation` (012)	194.2M	3.53M	3.89M	3.89M
`ic.pipeline` (001)	228.1M	37.96M	169.47M	**9.86M**

2184 stopped each child from parsing the whole 4.7 MB curve registry; a child now parses its one entry (190M instructions, about 98% of a rho or BSGS child). This PR stops each IC child from parsing the whole 2.8 MB alias table to match one curve name against the 25-row frozen calibration table. That cost 27M instructions at `0f5d9c272` and 154M on `main`. Each name is now resolved by binary search over the sorted text, with answers equal to the map's on every name (`curve_id::tests`).

Wall time per child (30 rounds, 2,000-resample round bootstrap)

comparison	inputs	ratio of medians	95% interval
after / before	all 18	**0.0786** (37.20 → 2.92 ms)	[0.0769, 0.0802]

after-aa / after (A/A)	all 18	1.0032	[0.9820, 1.0288]
this PR / `main`	all 18	0.9910	[0.9754, 1.0083]
this PR / `main`	`ic.pipeline` 001	**0.1457** (31.65 → 4.61 ms)	[0.1403, 0.1534]
this PR / `main`	`ic.pipeline` 010	**0.1400** (30.82 → 4.31 ms)	[0.1322, 0.1499]
`main` / after	all 18	1.1460	[1.1278, 1.1685]

The pooled A/A holds. Per input, host noise is about ±6%: one BSGS input's A/A reads 0.94 [0.89, 0.99]. The IC ratios are far outside that. `main` is about 15% slower than `0f5d9c272` on rho and BSGS children, with about 9% more instructions, from the measurement-hygiene work merged since (allocator trim, per-solve observations). That is not addressed here. Still spent before a rho solve under this PR (about 3.2M instructions): the registry entry search (1.2M, a linear scan because the registry is not sorted by slug), the argument parser (0.8M) and the curve's ICV1 identity (0.35M). The solve itself is about 1.1M.